

Inference for Set-Identified Lewbel Bounds in the SDF-News Notation

Rewritten using the notation of `lewbel_multivariate_set_identification.tex`

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Executive summary

This is the same substantive report as before, but rewritten in the notation of the attached TeX file. The key changes are notational and organizational:

1. The structural coefficient on the endogenous news vector is now $\theta \in \mathbb{R}^I$. A generic candidate value is $w \in \mathbb{R}^I$. In the one-endogenous-regressor case, set $I = 1$, so θ and w are scalars.
2. The common conditioning vector is X_t , the outcome is $Y_{1,t+1}$, and the endogenous news regressors are $Y_{2,t+1}$.

3. The residuals are $W_{1,t+1}$ and $W_{2,t+1}$, formed after projecting both equations on the same X_t .
4. The instrument list is $\mathcal{Z} = \{Z_{1,t+1}, \dots, Z_{J_Z,t+1}\}$, with imposed component-instrument pairs $(i, k) \in \mathcal{P}$.
5. The set-identified object is $\Theta(\mathcal{Z}, \tau)$. In the scalar case $I = 1$, this becomes the interval $[\underline{\theta}, \bar{\theta}]$. I avoid using L and U for the endpoints because the TeX notation already uses L_{ik} for a covariance moment.
6. For every candidate w , the other coefficients are recovered by the affine map

$$\beta_1(w) = \beta_1^R - (\beta_2^R)^\top w.$$

Thus, in the scalar case, once $[\underline{\theta}, \bar{\theta}]$ is known, the set for every component of β_1 is the image of this interval under that affine map.

The main inference conclusion is unchanged. Lewbel (2012) gives the population set and a plug-in way to estimate the set. The attached Baum-Lewbel advice paper and the standard Lewbel software cover the point-identified estimator, not confidence intervals for the set-identified Theorem 3 bounds. For the scalar triangular case, use a full bootstrap for endpoint uncertainty and then report either an Imbens-Manski/Stoye-style confidence interval for the true θ or a confidence set obtained by moment-inequality inversion.

1. Notation used in this report

The attached TeX file defines the common conditioning vector as

$$X_t := \begin{cases} (1, \text{PC}_t^\top)^\top, & H = 0, \\ (1, \text{PC}_t^\top, Y_{1,t}, \dots, Y_{1,t+1-H})^\top, & H \geq 1. \end{cases}$$

The triangular system is

$$Y_{1,t+1} = X_t^\top \beta_1 + \theta^\top Y_{2,t+1} + \varepsilon_{1,t+1}, \quad \theta \in \mathbb{R}^I,$$

$$Y_{2,t+1} = \beta_2^R X_t + \varepsilon_{2,t+1}, \quad \beta_2^R \in \mathbb{R}^{I \times \dim(X)}.$$

For the original one-endogenous-regressor question, take $I = 1$. Then $Y_{2,t+1}$, θ , w , and $\varepsilon_{2,t+1}$ are scalars. The notation still uses the same objects because the scalar case is just a special case of the attached TeX formulation.

The residualized variables are

$$\beta_1^R := (\mathbb{E}[X_t X_t^\top])^{-1} \mathbb{E}[X_t Y_{1,t+1}], \quad \beta_2^R := \mathbb{E}[Y_{2,t+1} X_t^\top] (\mathbb{E}[X_t X_t^\top])^{-1},$$

$$W_{1,t+1} := Y_{1,t+1} - X_t^\top \beta_1^R, \quad W_{2,t+1} := Y_{2,t+1} - \beta_2^R X_t.$$

Under $\mathbb{E}[X_t \varepsilon_{1,t+1}] = 0$ and $\mathbb{E}[X_t \varepsilon_{2,t+1}^\top] = 0$, the residualized model is

$$W_{1,t+1} = \theta^\top W_{2,t+1} + \varepsilon_{1,t+1}, \quad W_{2,t+1} = \varepsilon_{2,t+1}.$$

This residualized system is the object used for both point identification and set identification.

2. Exact Lewbel point identification in the attached notation

Let $Z_{k,t+1}$ be one scalar instrument from $\mathcal{Z}$. For each imposed pair $(i, k) \in \mathcal{P}$, define

$$L_{ik} := \text{Cov}(Z_{k,t+1}, W_{1,t+1}W_{2,i,t+1}),$$

$$Q_{ik} := \text{Cov}(Z_{k,t+1}, W_{2,t+1}W_{2,i,t+1}) \in \mathbb{R}^I,$$

$$P_{ik} := \text{Cov}(Z_{k,t+1}, W_{2,i,t+1}^2).$$

The exact Lewbel condition is the zero-contamination case

$$\text{Cov}(Z_{k,t+1}, \varepsilon_{1,t+1}\varepsilon_{2,i,t+1}) = 0.$$

Since the candidate first-equation error is

$$\varepsilon_{1,t+1}(w) := W_{1,t+1} - w^\top W_{2,t+1},$$

the exact condition implies

$$Q_{ik}^\top w = L_{ik}.$$

Stacking the rows $Q_{ik}^\top$ into $\mathcal{Q}$ and the scalars L_{ik} into $\mathcal{L}$ gives

$$\mathcal{Q}w = \mathcal{L}.$$

If $\text{rank}(\mathcal{Q}) = I$, the exact Lewbel system point-identifies

$$\theta = (\mathcal{Q}^\top \mathcal{Q})^{-1} \mathcal{Q}^\top \mathcal{L}.$$

In the single-news-component case $I = 1$, a single relevant exact instrument gives

$$\theta = \frac{L_{1k}}{Q_{1k}}, \quad Q_{1k} = P_{1k} = \text{Cov}(Z_{k,t+1}, W_{2,t+1}^2) \neq 0.$$

The standard Lewbel software implements this point-identified logic by estimating generated instruments and then running IV/GMM. It does not implement the set-identified bounds below.

3. Set identification with bounded relative correlations

The attached TeX file relaxes exact validity by imposing, for every $(i, k) \in \mathcal{P}$,

$$|\text{Corr}(Z_{k,t+1}, \varepsilon_{1,t+1}\varepsilon_{2,i,t+1})| \leq \tau_{ik} |\text{Corr}(Z_{k,t+1}, \varepsilon_{2,i,t+1}^2)|, \quad 0 \leq \tau_{ik} < 1.$$

The scalar τ_{ik} is a sensitivity parameter. It is not identified from the data without extra assumptions. The special case $\tau_{ik} = 0$ is exact Lewbel validity.

For each news component i , define the product-variance moments

$$S_i^{(0)} := \text{Var}(W_{1,t+1}W_{2,i,t+1}),$$

$$S_i^{(1)} := \text{Cov}(W_{2,t+1}W_{2,i,t+1}, W_{1,t+1}W_{2,i,t+1}) \in \mathbb{R}^I,$$

$$S_i^{(2)} := \text{Var}(W_{2,t+1}W_{2,i,t+1}) \in \mathbb{R}^{I \times I}, \quad \sigma_i^2 := \text{Var}(W_{2,i,t+1}^2) > 0.$$

For any candidate w ,

$$\text{Cov}(Z_{k,t+1}, \varepsilon_{1,t+1}(w)W_{2,i,t+1}) = L_{ik} - Q_{ik}^\top w,$$

$$\text{Var}(\varepsilon_{1,t+1}(w)W_{2,i,t+1}) = S_i^{(0)} - 2S_i^{(1)\top}w + w^\top S_i^{(2)}w.$$

After squaring the relative-correlation bound and cancelling $\text{Var}(Z_{k,t+1})$, each pair (i, k) supplies the quadratic inequality

$$g_{ik}(w) := (L_{ik} - Q_{ik}^\top w)^2 - d_{ik}(S_i^{(0)} - 2S_i^{(1)\top}w + w^\top S_i^{(2)}w) \leq 0,$$

where

$$d_{ik} := \tau_{ik}^2 \frac{P_{ik}^2}{\sigma_i^2}.$$

Equivalently,

$$g_{ik}(w) = w^\top A_{ik}w + b_{ik}^\top w + c_{ik} \leq 0,$$

$$A_{ik} = Q_{ik}Q_{ik}^\top - d_{ik}S_i^{(2)}, \quad b_{ik} = -2L_{ik}Q_{ik} + 2d_{ik}S_i^{(1)}, \quad c_{ik} = L_{ik}^2 - d_{ik}S_i^{(0)}.$$

The identified set for the structural news coefficients is

$$\Theta(\mathcal{Z}, \tau) := \{w \in \mathbb{R}^I : g_{ik}(w) \leq 0 \text{ for every } (i, k) \in \mathcal{P}\}.$$

Under the maintained assumptions in the attached TeX file, $\theta \in \Theta(\mathcal{Z}, \tau)$. The set can be empty if the model and the asserted slack values are mutually inconsistent, and it can be unbounded if the imposed instruments do not restrict some directions in $\mathbb{R}^I$.

3.1 Scalar specialization: the original one-endogenous-regressor case

For $I = 1$, suppress $i = 1$ and write $w \in \mathbb{R}$. For instrument k , the pair-specific inequality is

$$g_k(w) = A_k w^2 + b_k w + c_k \leq 0,$$

where

$$A_k = Q_k^2 - d_k S^{(2)}, \quad b_k = -2L_k Q_k + 2d_k S^{(1)}, \quad c_k = L_k^2 - d_k S^{(0)}, \quad d_k = \tau_k^2 \frac{P_k^2}{\sigma^2}.$$

In the scalar case,

$$Q_k = P_k = \text{Cov}(Z_{k,t+1}, W_{2,t+1}^2), \quad S^{(2)} = \sigma^2 = \text{Var}(W_{2,t+1}^2),$$

so, with $0 \leq \tau_k < 1$ and $P_k \neq 0$, $A_k = (1 - \tau_k^2)P_k^2 > 0$. Therefore the single-pair set is a closed interval bounded by the roots of $g_k(w) = 0$:

$$\underline{\theta}_k = \frac{-b_k - \sqrt{b_k^2 - 4A_k c_k}}{2A_k}, \quad \bar{\theta}_k = \frac{-b_k + \sqrt{b_k^2 - 4A_k c_k}}{2A_k}.$$

With multiple scalar instruments, the plug-in scalar identified set is the intersection of the pairwise intervals:

$$\Theta(\mathcal{Z}, \tau) = \bigcap_{k:(1,k) \in \mathcal{P}} [\underline{\theta}_k, \bar{\theta}_k].$$

When the intersection is nonempty,

$$\underline{\theta} = \max_k \underline{\theta}_k, \quad \bar{\theta} = \min_k \bar{\theta}_k.$$

When the intersection is empty, the maintained model and the chosen slack profile τ are refuted by the imposed restrictions.

4. What the attached papers provide and what must be added

The papers cover the estimation problem only up to the plug-in set. They do not give a complete inference procedure for the set-identified Theorem 3 case.

Object	Status in Lewbel (2012) and Baum-Lewbel (2019)	What this report adds
Point-identified Lewbel estimator	Covered by 2SLS/GMM generated-instrument logic	No special partial-identification issue
Point-identified standard errors	Covered by standard IV/GMM asymptotics, with generated-instrument details	No change
Set $\Theta(\mathcal{Z}, \tau)$	Population characterization supplied by Lewbel's Theorem 3 and generalized in the attached TeX notation	Restated in $L_{ik}, Q_{ik}, P_{ik}, S_i^{(j)}$ notation
Plug-in set estimator	Suggested by replacing population moments by sample moments	Detailed algorithm below
Endpoint standard errors for $\hat{\underline{\theta}}, \hat{\bar{\theta}}$	Not supplied	Bootstrap or delta-method recommendations
Confidence set for the population identified set	Not supplied	Conservative outer set or joint endpoint bootstrap
Confidence interval for true scalar θ	Not supplied	Imbens-Manski/Stoye-style interval or moment-inequality inversion

Object	Status in Lewbel (2012) and Baum-Lewbel (2019)	What this report adds
Inference for $\beta_1(w)$	Not supplied	Affine mapping and bootstrap/projection recommendations
Software for Theorem 3 set inference	No dedicated package located	Custom code plus general moment-inequality tools

5. Sample construction of the plug-in set

Use the common trimmed sample implied by the lag order H . With the $1/T$ normalization,

$$\widehat{\text{Cov}}(a, b) := T^{-1} \sum_t a_t b_t^\top - \bar{a} \bar{b}^\top, \quad \widehat{\text{Var}}(a) := \widehat{\text{Cov}}(a, a),$$

the plug-in algorithm is:

1. Regress $Y_{1,t+1}$ on the full common vector X_t . Save $\hat{\beta}_1^R$ and residuals $\widehat{W}_{1,t+1}$.
2. Regress every component of $Y_{2,t+1}$ on the same X_t . Save $\hat{\beta}_2^R$ and residuals $\widehat{W}_{2,t+1}$.
3. Construct the pre-specified instruments $Z_{k,t+1}$ or the date- t specialization $Z_{k,t}$, and specify $\mathcal{P}$.
4. Compute sample analogues of L_{ik} , Q_{ik} , P_{ik} , $S_i^{(0)}$, $S_i^{(1)}$, $S_i^{(2)}$, and σ_i^2 .
5. Build $\hat{g}_{ik}(w)$ and solve the quadratically constrained problems defining $\widehat{\Theta}(\mathcal{Z}, \tau)$.

For $I = 1$, step 5 reduces to solving quadratic roots for each scalar instrument and intersecting the resulting intervals.

6. The three different inference targets

The most important practical point is that there are three different objects. They should not be reported as if they were the same.

6.1 Endpoint uncertainty

In the scalar case, the plug-in endpoints $\hat{\underline{\theta}}$ and $\hat{\bar{\theta}}$ are sample statistics. Their standard errors describe sampling variation in the estimated lower and upper bounds. They do not by themselves give a confidence interval for the true θ .

A practical full bootstrap is:

1. Resample observations. For time-series data, use a block bootstrap; for clustered data, resample clusters.
2. In each resample, recompute $\hat{\beta}_1^R$, $\hat{\beta}_2^R$, $\widehat{W}_{1,t+1}$, and $\widehat{W}_{2,t+1}$.
3. Recompute all sample analogues $\hat{L}_{ik}$, $\hat{Q}_{ik}$, $\hat{P}_{ik}$, $\hat{S}_i^{(j)}$, and $\hat{\sigma}_i^2$.
4. Recompute $\widehat{\Theta}^*(\mathcal{Z}, \tau)$. In the scalar case, store $\hat{\underline{\theta}}^*$ and $\hat{\bar{\theta}}^*$.
5. Use the empirical standard deviations of the bootstrap endpoint draws as $\widehat{\text{se}}(\hat{\underline{\theta}})$ and $\widehat{\text{se}}(\hat{\bar{\theta}})$.

The bootstrap should include failures as diagnostics. Frequent failures can signal weak heteroskedastic relevance $\hat{P}_{ik} \approx 0$, a near-zero discriminant, nonempty-set fragility, or a slack value that is too small for the maintained restrictions.

6.2 Confidence set for the identified interval

A confidence set for the population identified interval $[\underline{\theta}, \bar{\theta}]$ is meant to contain the entire interval. A simple conservative construction is

$$CS_{\Theta, 1-\alpha}^{\text{outer}} = \left[\hat{\underline{\theta}} - z_{1-\alpha/2} \widehat{\text{se}}(\hat{\underline{\theta}}), \hat{\bar{\theta}} + z_{1-\alpha/2} \widehat{\text{se}}(\hat{\bar{\theta}}) \right].$$

A bootstrap analogue can use one-sided quantiles of the endpoint errors instead of normal critical values. This object answers the question: where is the whole population identified set? It is generally wider than an interval designed only to cover the true scalar θ .

6.3 Confidence interval for the true scalar coefficient θ

A confidence interval for the true coefficient covers the unknown scalar θ , which is known only to lie somewhere inside $[\underline{\theta}, \bar{\theta}]$. This is the Imbens-Manski/Stoye problem.

Let

$$s_\ell := \widehat{\text{se}}(\hat{\underline{\theta}}), \quad s_u := \widehat{\text{se}}(\hat{\bar{\theta}}).$$

A common practical critical value c solves

$$\Phi \left(c + \frac{\max(\hat{\bar{\theta}} - \hat{\underline{\theta}}, 0)}{\max(s_\ell, s_u)} \right) - \Phi(-c) = 1 - \alpha.$$

Then report

$$CI_{\theta, 1-\alpha}^{\text{IM}} = \left[\hat{\underline{\theta}} - cs_\ell, \hat{\bar{\theta}} + cs_u \right].$$

When the identified interval is wide relative to sampling error, c is close to $z_{1-\alpha}$. When the identified interval collapses toward a point, c approaches $z_{1-\alpha/2}$. In nonregular cases - for example nearly equal roots, weak relevance, or multiple instruments where the active endpoint changes across bootstrap samples - moment-inequality inversion is safer.

6.4 Moment-inequality inversion

The bounded-correlation restriction can be used directly as a moment-inequality test. For a candidate w , retain it if the sample analogue of

$$g_{ik}(w) \leq 0 \quad \text{for all } (i, k) \in \mathcal{P}$$

is not rejected.

In the scalar case, this means testing values $w \in \mathbb{R}$ on a grid or by root-finding. A simple max statistic is

$$T(w) = \max_{(i,k) \in \mathcal{P}} \frac{\sqrt{T} \hat{g}_{ik}(w)}{\hat{s}_{ik}(w)},$$

where $\hat{s}_{ik}(w)$ is a standard error for $\hat{g}_{ik}(w)$. The confidence set is

$$CS_{\theta,1-\alpha}^{\text{MI}} = \{w : T(w) \leq c_{1-\alpha}(w)\},$$

with $c_{1-\alpha}(w)$ obtained by bootstrap, multiplier bootstrap, or generalized moment-selection methods. This is the most defensible route when the endpoints are nonregular or when the model has several imposed instruments and the active inequalities can change.

One can also test the two primitive one-sided restrictions

$$\text{Corr}(Z_{k,t+1}, \varepsilon_{1,t+1}(w)W_{2,i,t+1}) - \tau_{ik} |\text{Corr}(Z_{k,t+1}, W_{2,i,t+1}^2)| \leq 0,$$

$$- \text{Corr}(Z_{k,t+1}, \varepsilon_{1,t+1}(w)W_{2,i,t+1}) - \tau_{ik} |\text{Corr}(Z_{k,t+1}, W_{2,i,t+1}^2)| \leq 0.$$

This avoids relying only on endpoint delta-method logic and keeps the inference aligned with the maintained inequality model.

7. Mapping inference to the other coefficients

For every candidate $w \in \Theta(\mathcal{Z}, \tau)$, the coefficient vector on X_t is uniquely determined by

$$\beta_1(w) = \beta_1^R - (\beta_2^R)^\top w.$$

Therefore the identified set for β_1 is

$$\mathcal{A} := \{\beta_1^R - (\beta_2^R)^\top w : w \in \Theta(\mathcal{Z}, \tau)\}.$$

For $I = 1$, this becomes the affine curve

$$\beta_1(w) = \beta_1^R - (\beta_2^R)^\top w, \quad w \in [\underline{\theta}, \bar{\theta}].$$

For component j , if $\beta_{2,j}^R$ denotes the coefficient of the j -th element of X_t in the scalar Y_2 -on- X_t projection, then

$$\beta_{1,j}(w) = \beta_{1,j}^R - \beta_{2,j}^R w.$$

The sharp plug-in bounds are

$$\left[\min\{\hat{\beta}_{1,j}(\hat{\underline{\theta}}), \hat{\beta}_{1,j}(\hat{\bar{\theta}})\}, \max\{\hat{\beta}_{1,j}(\hat{\underline{\theta}}), \hat{\beta}_{1,j}(\hat{\bar{\theta}})\} \right].$$

For uncertainty, use the same bootstrap draws used for θ , recomputing $\hat{\beta}_1^{R*}$, $\hat{\beta}_2^{R*}$, and the endpoint images in every draw. Componentwise beta intervals are not a joint confidence band for all components. For a joint statement about (θ, β_1) , use projection or test inversion.

8. Practical reporting template in the attached notation

A credible empirical report should include:

1. The common conditioning vector X_t , including the principal components and lag choices.
2. The instrument list $\mathcal{Z}$, the pair set $\mathcal{P}$, and whether each Z_k is date- t or date- $(t + 1)$ adapted.
3. A grid of slack values τ , for example 0, 0.10, 0.25, 0.50, and 0.90, unless a specific slack is justified externally.
4. Relevance diagnostics: $\widehat{P}_{ik} = \widehat{\text{Cov}}(Z_{k,t+1}, \widehat{W}_{2,i,t+1}^2)$ and, for exact point-identified Lewbel estimates, generated-instrument first-stage diagnostics.
5. The plug-in scalar interval $[\underline{\theta}, \hat{\theta}]$ when $I = 1$, or coordinate projections $[\underline{\theta}_m, \hat{\theta}_m]$ when $I > 1$.
6. Endpoint standard errors from the full bootstrap.
7. A confidence interval for the true scalar θ , preferably Imbens-Manski/Stoye-style or moment-inequality inversion.
8. A separate outer confidence set for the whole identified set only if that is the target.
9. The mapped coefficient set $\widehat{\mathcal{A}} = \{\hat{\beta}_1^R - (\hat{\beta}_2^R)^\top w : w \in \widehat{\Theta}(\mathcal{Z}, \tau)\}$.
10. Bootstrap failures, empty-set draws, no-real-root draws, and any numerical nonconvergence.

Important cautions:

- τ_{ik} is a maintained sensitivity bound, not a parameter learned from the model.
- Data-adaptive choice of $\mathcal{Z}$ requires sample splitting or a uniform inference correction.
- Weak heteroskedastic relevance, $\widehat{P}_{ik} \approx 0$, makes both the set and the inference unstable.
- Generated Lewbel instruments should not be squared, interacted, or transformed without additional assumptions.
- The Lewbel estimator targets homogeneous linear structural coefficients; it is not automatically a local-average-treatment-effect estimator.

9. Software landscape

The software conclusion is the same as in the previous report, updated here in the attached notation.

Need	R	Stata	Comments
Point-identified Lewbel estimator	<code>REndo::hetErrorsIV;</code> older <code>ivlewbelt::lewbelt</code>	<code>ivreg2h</code>	These estimate point-identified generated-instrument IV/GMM models and standard errors.
Plug-in $\widehat{\Theta}(\mathcal{Z}, \tau)$ for Lewbel's set case	No standard package found	No dedicated command found	Easy to code for $I = 1$; quadratically constrained optimization needed for $I > 1$.
Confidence intervals for Lewbel Theorem 3 set bounds	No standard package found	No dedicated command found	Requires custom bootstrap, Imbens-Manski/Stoye logic, or moment-inequality inversion.
General moment-inequality inference	No widely used drop-in CRAN package found for this exact task	<code>cmtest</code> , <code>clrbound</code>	Useful templates, not turnkey Lewbel-set commands.

Need	R	Stata	Comments
GMM building blocks	gmm, momentfit	built-in/user-written GMM	Useful for custom moments, not partial-ID inference by themselves.

Validation notes checked during this rewrite:

- The R package REndo manual describes hetErrorsIV as implementing Lewbel’s 2012 heteroskedasticity approach and lists package version 2.5.0 on r-universe/CRAN infrastructure.
- The older R package ivlewbels documents a lewbel function for heteroskedasticity-based GMM estimation, but it is old and is not a Theorem 3 bound-inference package.
- The Stata command ivreg2h is described on RePEc/SSC as generating Lewbel-style instruments and estimating IV regressions; it is a point-identified IV/GMM implementation, not a set-bound package.
- The Stata cmittest and clrbound commands implement general conditional moment-inequality/intersection-bound methods, but the Lewbel inequalities $g_{ik}(w) \leq 0$ must be supplied or adapted by the user.

10. Minimal R template in the attached notation

The following template implements the scalar case $I = 1$. It uses the notation from the attached TeX file: θ is the scalar coefficient, the endpoints are $\underline{\theta}$ and $\bar{\theta}$, and the recovery map is $\beta_1(w) = \beta_1^R - (\beta_2^R)^\top w$. The code accepts one or more scalar instruments in the columns of Z and intersects the resulting intervals.

```

lewbel_theta_set <- function(y1, y2, X, Z, tau = 0.5, tol = 1e-10) {
  y1 <- as.numeric(y1)
  y2 <- as.numeric(y2)
  X <- as.matrix(X)
  Z <- as.matrix(Z)

  n <- length(y1)
  stopifnot(length(y2) == n, nrow(X) == n, nrow(Z) == n)

  K <- ncol(Z)
  if (length(tau) == 1L) tau <- rep(tau, K)
  stopifnot(length(tau) == K, all(tau >= 0), all(tau < 1))

  XtX <- crossprod(X)
  beta1_R <- as.numeric(qr.solve(XtX, crossprod(X, y1)))
  beta2_R <- as.numeric(qr.solve(XtX, crossprod(X, y2)))

  W1 <- as.numeric(y1 - X %*% beta1_R)
  W2 <- as.numeric(y2 - X %*% beta2_R)

  cv <- function(a, b) mean((a - mean(a)) * (b - mean(b)))
  vv <- function(a) mean((a - mean(a))^2)

  pair_intervals <- matrix(NA_real_, K, 2,
    dimnames = list(colnames(Z), c("theta_lower", "theta_upper")))
  pair_moments <- matrix(NA_real_, K, 8,
    dimnames = list(colnames(Z),
      c("L", "Q", "P", "S0", "S1", "S2",

```

```

"discriminant", "cov_Z_W2sq"))))

for (k in seq_len(K)) {
  z <- Z[, k]
  L <- cv(z, W1 * W2)
  Q <- cv(z, W2^2)
  P <- Q
  S0 <- vv(W1 * W2)
  S1 <- cv(W2^2, W1 * W2)
  S2 <- vv(W2^2)
  sigma2 <- S2

  if (abs(P) < tol) stop("Weak/no relevance: Cov(Z_k, W2^2) is near zero.")
  if (sigma2 < tol) stop("Var(W2^2) is near zero.")

  d <- tau[k]^2 * P^2 / sigma2
  A <- Q^2 - d * S2
  b <- -2 * L * Q + 2 * d * S1
  c0 <- L^2 - d * S0

  disc <- b^2 - 4 * A * c0
  if (disc < -tol) stop("No real roots for at least one instrument/slack pair.")
  disc <- max(disc, 0)

  roots <- sort(c((-b - sqrt(disc)) / (2 * A),
                  (-b + sqrt(disc)) / (2 * A)))

  pair_intervals[k, ] <- roots
  pair_moments[k, ] <- c(L, Q, P, S0, S1, S2, disc, P)
}

theta_lower <- max(pair_intervals[, "theta_lower"])
theta_upper <- min(pair_intervals[, "theta_upper"])
empty <- theta_lower > theta_upper

betal_at <- function(w) betal_R - beta2_R * w
betal_bounds <- if (!empty) {
  B <- cbind(lower_endpoint = betal_at(theta_lower),
             upper_endpoint = betal_at(theta_upper))
  out <- t(apply(B, 1, range))
  colnames(out) <- c("betal_lower", "betal_upper")
  out
} else {
  NULL
}

list(
  theta = c(theta_lower = theta_lower, theta_upper = theta_upper),
  empty = empty,
  betal_R = betal_R,
  beta2_R = beta2_R,
  betal_bounds = betal_bounds,
  pair_intervals = pair_intervals,
  pair_moments = pair_moments
)
}

```

```

im_critical <- function(width, se_lower, se_upper, alpha = 0.05) {
  s <- max(se_lower, se_upper)
  if (!is.finite(s) || s <= 0) return(qnorm(1 - alpha))
  delta <- max(width, 0) / s
  f <- function(c) pnorm(c + delta) - pnorm(-c) - (1 - alpha)
  uniroot(f, interval = c(0, 10))$root
}

lewbelt_theta_set_boot <- function(y1, y2, X, Z, tau = 0.5,
                                   B = 999, alpha = 0.05, seed = NULL) {
  if (!is.null(seed)) set.seed(seed)
  base <- lewbelt_theta_set(y1, y2, X, Z, tau)
  n <- length(y1)

  boot_theta <- matrix(NA_real_, B, 2,
                      dimnames = list(NULL, c("theta_lower", "theta_upper")))
  failures <- 0L

  for (b in seq_len(B)) {
    idx <- sample.int(n, n, replace = TRUE)
    out <- try(
      lewbelt_theta_set(y1[idx], y2[idx], X[idx, , drop = FALSE], Z[idx, , drop = FALSE], tau),
      silent = TRUE
    )
    if (inherits(out, "try-error") || isTRUE(out$empty)) {
      failures <- failures + 1L
    } else {
      boot_theta[b, ] <- out$theta
    }
  }

  ok <- complete.cases(boot_theta)
  se <- apply(boot_theta[ok, , drop = FALSE], 2, sd)
  zcrit <- qnorm(1 - alpha / 2)

  outer_set <- c(
    theta_lower = base$theta["theta_lower"] - zcrit * se["theta_lower"],
    theta_upper = base$theta["theta_upper"] + zcrit * se["theta_upper"]
  )

  c_im <- im_critical(diff(base$theta), se["theta_lower"], se["theta_upper"], alpha)
  ci_theta <- c(
    theta_lower = base$theta["theta_lower"] - c_im * se["theta_lower"],
    theta_upper = base$theta["theta_upper"] + c_im * se["theta_upper"]
  )

  list(
    estimate = base,
    boot_theta = boot_theta,
    se_theta = se,
    outer_confidence_set_for_Theta = outer_set,
    im_critical = c_im,
    confidence_interval_for_theta = ci_theta,
    bootstrap_successes = sum(ok),
    bootstrap_failures = failures
  )
}

```

```
)
}
```

For time-series applications, replace the row bootstrap in `lewbel_theta_set_boot()` with a block bootstrap. For cluster samples, resample clusters.

11. Bottom line

Using the attached notation, the final practical recommendation is:

1. Estimate $\widehat{W}_{1,t+1}$ and $\widehat{W}_{2,t+1}$ from projections on the common X_t .
2. For pre-specified $\mathcal{Z}$, $\mathcal{P}$, and τ , compute $\widehat{\Theta}(\mathcal{Z}, \tau)$ from the inequalities $\hat{g}_{ik}(w) \leq 0$.
3. In the scalar case $I = 1$, report $[\hat{\theta}, \hat{\theta}]$, obtained as the intersection of quadratic-root intervals.
4. Bootstrap the entire construction for endpoint uncertainty.
5. Report an Imbens-Manski/Stoye-style confidence interval for the true scalar θ , or use moment-inequality inversion.
6. Recover and report the corresponding set for the other equation coefficients with

$$\hat{\beta}_1(w) = \hat{\beta}_1^R - (\hat{\beta}_2^R)^\top w.$$

No standard package appears to implement the full Lewbel set-bound inference procedure in this notation. The point-identified Lewbel estimator is available in R and Stata, but the set-identified case requires custom code or adaptation of general moment-inequality tools.

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